

2. LangGraph

LangGraph is an open-source library built by LangChain used to create complex, multi-agent AI workflows structured as state machines. [\[1, 2, 3, 4, 5\]](#)

- **The Problem:** Standard LLM chains move strictly in one direction (Step A → Step B → End). They cannot easily handle loops, error correction, or autonomous decision-making. [\[1, 2, 3, 4, 5\]](#)
- **The Solution:** LangGraph models workflows as **Graphs** consisting of **Nodes** (Python functions or LLM calls) and **Edges** (conditional routing paths). [\[1, 2, 3, 4, 5\]](#)
- **The Result:** It allows you to build "Agentic" systems. For example, if an LLM writes bad code, LangGraph can route that code to a tester node, catch the error, loop back to the LLM to fix it, and repeat until the code passes validation